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Long-term treatment outcome and mutational analysis of patients on third-line antiretroviral therapy in programmatic conditions

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Background: In low- and middle-income countries where most patients receive standardized third-line ART through national programmes, real-world data are scarce. This study was done to assess the long-term survival, and virological and mutational outcomes of people living with HIV receiving third-line ART between July 2016 and December 2019 in an ART centre in India.

Methods: Eighty-five patients were started on third-line ART. Genotypic resistance testing to identify drug resistance mutations in the integrase, reverse transcriptase and protease genes was done at the start of third-line therapy, as well as in those who did not attain virological suppression after 12 months of therapy.

Results: Survival was 85% (72/85) at 12 months and 72% (61/85) at the end of follow-up in March 2022. Virological suppression was present in 82% (59/72) and 88% (59/67) at 12 months and at the end of follow-up, respectively. Five out of 13 patients who had virological failure at 12 months showed virological suppression at the end of the study. At the start of third-line therapy, 35% (14/40) and 45% (17/38) of patients had major integrase- and protease-associated mutations, respectively, even though they had never been on integrase inhibitor-based regimens. At 1 year follow-up, among those failing third-line therapy, 33% (4/12) of patients had major integrase mutations, but none had major protease mutations.

Conclusions: This study demonstrates good long-term outcome in patients on standardized third-line ART in programmatic conditions with very few mutations in those failing the therapy.

Introduction

India's ART programme is the world's second largest, and it has been recognized as one of the best public health programmes for care of people living with HIV (PLHIV). Programmatic achievements of India have made a significant contribution to global progress in the fight against HIV/AIDS. There is direct access to diagnostic facilities, free first-, second- and third-line ART for PLHIV in the national programme.¹ As of March 2021, there were around 1.49 million PLHIV on ART in India, of which 1.38 million were receiving free ART from the 619 ART clinics run by the National AIDS Control Program (NACP). Around 65 000 people were on second-line ART, and about 3000 were on third-line ART.²

NACP has a multi-layered service delivery method in which third-line therapy is provided by 11 centres of excellence (CoEs) in HIV care in the country since May 2016.² The programme adopted the WHO recommendations for constituting a standardized third-line regimen and a combination of darunavir/ritonavir and raltegravir was provided for those failing second-line treatment since May 2016.³ Raltegravir was replaced by dolutegravir in 2020.¹ Genotyping is not routinely done in the NACP.

In low- and middle-income countries where most patients receive standardized third-line ART through national programmes, real-world data are scarce on survival, virological outcomes and drug resistance mutations (DRMs) for those failing therapy.^{4–9} Mutational data on HIV-1 subtype C (the commonest in low- and middle-income countries) in patients who have failed an integrase inhibitor-based regimen are also limited.⁵ Thus, in this study, we have evaluated the long-term virological and survival outcome of PLHIV on third-line ART in the NACP and also assessed the DRMs at baseline as well as at the time of treatment failure at 1 year.

Materials and methods

This single-centre observational study was done at a CoE in HIV care, Banaras Hindu University (BHU) in a tertiary hospital in Varanasi, India. CoE, BHU has one of the largest ART centres in the eastern part of the country, serving 4550 PLHIV from states of Uttar Pradesh, Bihar, Madhya Pradesh and Jharkhand. The study was approved by the Ethics Committee of the Institute of Medical Sciences, Banaras Hindu University (No. Dean/2015-16/EC/413 and No. Dean/2018/EC/715).

Study population

Patients failing second-line therapy from the National AIDS Control Organization (NACO)-funded ART centres of Uttar Pradesh and Bihar are referred to the State AIDS Clinical Expert Panel (SACEP) in CoE BHU for initiation of third-line therapy.

All PLHIV who were started on third-line ART between July 2016 and December 2019 in the ART centre were included in the study. The criteria for starting third-line therapy were patients on second-line therapy with plasma viral load (VL) > 1000 copies/mL on two occasions.

Information on demographics, clinical status, opportunistic infection, previous ART regimen, CD4 count, VL and other laboratory parameters were abstracted from white card available at the ART centre.

VL was done on the HIV viral load assay platform Abbot Molecular System (detection level 150 copies/mL) at 6 monthly intervals as per guidelines. Patients with VL > 1000 copies/mL were put on step-up adherence counselling and VL was repeated at 3 months. Virological failure was defined as plasma VL > 1000 copies/mL on two occasions. Genotypic resistance testing to identify DRMs in the integrase (IN), reverse transcriptase (RT) and protease (PR) genes was done at the start of third-line therapy as well as in those who did not attain virological suppression after 12 months of therapy. Drug regimens were not modified according to the genotyping results. Patients visited the CoE every month to collect their medication. Those patients who did not come for three consecutive months at the ART centre were labelled as lost to follow-up (LFU) at the end of the fourth month, as per NACO guidelines. Adherence was calculated on the basis of pill count at every visit. For analysis we compared those with >95% adherence at every visit with those with <95% at any visit.

Patients were routinely followed up in the ART centre and long-term survival and virological outcome was assessed in March 2022.

Sample collection and processing

Whole blood was collected from patients into EDTA tubes and used for CD4 estimation; in parallel, plasma was separated following centrifugation at 2000 rpm at 4°C for 10 min and stored at -80°C until viral RNA isolation. The CD4 T cell counts for all these patients were performed using the BD FACSCount system (Becton, Dickinson).

Nucleic acid extraction

The HIV RNA was extracted manually using a QIAamp® Viral RNA mini kit according to the manufacturer's instructions (QIAGEN, Germany; Cat No. 52906). Viral RNA was stored at -80°C until use.

Amplification of RT and PR genes

RT-PCR

Amplification of complete PR (codons 1 to 99) and partial RT (codons 1 to 256) regions of the *pol* gene was performed by a previously published method.^{10,11} Synthesis of cDNA and PCR was performed using the ThermoScript RT PCR System (Invitrogen; Cat No. 1146016) with Platinum Taq DNA polymerase (Invitrogen; Cat No. 10966018). cDNA was synthesized in a 20 µL reaction mixture using ThermoScript RT enzyme and primer (5'-GGATAAATCTGACTTGCCART-3', HxB2 position; 3343). The thermal cycling conditions were 65°C for 5 min for denaturation followed by RT incubation (42°C for 15 min and 50°C for 15 min). After cDNA was synthesized, we used Platinum Taq polymerase enzyme and two outer primers (5'-GGATAAATCTGACTTGCCART-3', HxB2 position 3343; and 5'-AGGAAG GACACCATGAARGA-3', HxB2 position 2042) for the first round of PCR using a 50 µL reaction mixture. Thermal cycling conditions for the first round of PCR were initial denaturation at 94°C for 2 min, followed by 30 cycles of 94°C for 20 s, 50°C for 20 s and 72°C for 60 s; the final elongation step was done at 72°C for 5 min.

Nested PCR

The second round of PCR was performed using a 50 µL volume of reaction mixture, the same as the first-round PCR reaction mixture and two inner primers (5'-GCTTCCCTCARATCACTCTT-3', HxB2 location 2248; and 5'-CCACTAAYT TCTGTA TRTCATTGAC-3', HxB2 location 3309) with an expected fragment size of 1086 bp. Amplification started with initial denaturation at 94°C for 2 min, followed by 30 cycles of denaturation at 94°C for 20 s, annealing at 50°C for 20 s and elongation at 72°C for 60 s. The final elongation step was done at 72°C for 5 min and products were stored at 4°C until further use.

HIV-1 integrase gene amplification

Amplification of the HIV-1 IN gene was carried out by following the method used by Sachithanandham *et al.*¹² in 2016.

Sequencing

Samples were processed after gel purification using the QIAquick gel extraction kit (Cat. No. 28704). Sequencing was done by a third-party commercial organization (Medauxin) using an Applied Biosystem 3730xl DNA Analyzer.

Sequence analysis and subtyping

Following sequencing, we used the Recall software to build a contig using <https://recall.bccfe.ca/> and the drug resistance pattern was determined by the Stanford University HIV Genotypic Resistance Interpretation Algorithm version 9.0 (<https://hivdb.stanford.edu/hivdb/by-mutations/>). HIV-1 subtyping was performed on all of the samples in accordance with the recommendations of the REGA HIV-1 subtyping tool version 3.0 (<http://dbpartners.stanford.edu:8080/RegaSubtyping/stanford-hiv/typingtool/>).

Statistical analysis

The statistical analysis was performed using Statistical Package for the Social Sciences (SPSS, Version 23.0; IBM Corp., NY, USA). Descriptive statistics were used—mean ± SD for quantitative variables, and frequency with percentage distribution for categorized variables. For categorical variables, chi-squared and Fischer's exact tests were used, and for comparing two groups of mean or median, Student's *t*-test and Mann-Whitney *U*-test were used. Probability of significance (*P* value) of ≤0.05 was considered as statistically significant. A multivariate analysis was done to assess the predictors of death in third-line patients using the binary logistic regression model, including all variables that were associated with the outcome (*P* < 0.05) in univariate analysis.

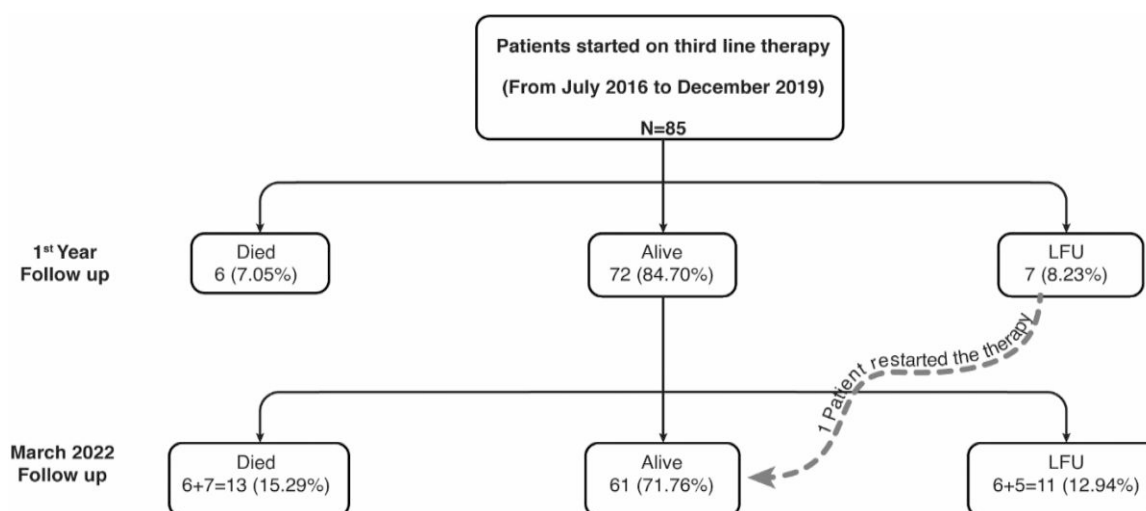
Results

Baseline characteristics

Eighty-seven patients were started on third-line ART between July 2016 and December 2019. Out of the 87 patients, 2 were initiated on third-line therapy from a private clinic and were not on the standardized third-line treatment recommended by the programme, thus only 85 patients were included in the analysis. The baseline characteristics (at the time of diagnosis of second-line failure) of the 85 patients initiating third-line ART are summarized in Table 1. The mean age at initiation of third-line ART was 40.98 ± 10.25 years, and 74% were male. The mean duration of first-line and second-line ART treatment was 56.52 ± 34.16 and 36.93 ± 22.37 months, respectively, and the mean duration of third-line ART was 42.08 ± 18.82 months. The median CD4 count

Table 1. Demographic details at the start of third-line therapy and risk factor analysis of surviving and dead patients at the end of follow-up of study

Characteristics	Patients (N=85)	Deceased (n=13)	Survived (n=61)	P value
Age (years), n (%)				
>40	45 (52.9)	6 (46.2)	34 (55.7)	0.529
<40	40 (47.1)	7 (53.8)	27 (44.3)	0.531
Mean±SD	40.98±10.25	39.46±10.89	41.49±10.48	
Sex, n (%)				
Male	63 (74.1)	8 (61.5)	48 (78.7)	0.191
Female	22 (25.9)	5 (38.5)	13 (21.3)	
WHO clinical stage, n (%)				
III+IV	4 (4.8)	3 (23.1)	1 (1.7)	0.002
I+II	80 (95.2)	10 (76.9)	59 (98.3)	
First-line duration (months), mean±SD	56.52±34.16	65.08±34.27	55.24±32.00	0.349
Second-line duration (months), mean±SD	36.93±22.37	34.31±25.23	39.02±22.49	0.504
Third-line duration (months), mean±SD	42.08±18.82	19.54±15.35	49.39±13.33	<0.001
Weight (kg), mean±SD	49.66±12.84	42.31±16.11	51.13±11.55	0.023
CD4 count (cells/mm ³), n (%)				
<350	58 (68.2)	8 (61.5)	42 (68.9)	0.609
>350	27 (31.8)	5 (38.5)	19 (31.1)	
Mean±SD	321.08±222.11	335±263.72	327.25±216.76	0.911
VL (copies/mL), median (IQR)	37 593(6870.50–194 876.50)	863 453±2 748 373	252 918±706 195	0.127
Log ₁₀ value	4.56	5.94	5.40	
Haemoglobin (g/dL), n (%)				
<12	39 (47.6)	9 (69.2)	30 (50.8)	0.229
>12	43 (52.4)	4 (30.8)	29 (49.2)	
Mean±SD	11.68±2.34	10.5±2.44	11.78±2.13	0.062
Adherence at 1 year, n (%)				
<95		7 (63.6)	21 (34.4)	0.067
>95		4 (36.4)	40 (65.6)	

**Figure 1.** Survival outcomes of patients started on third-line therapy.

and VL at third-line ART initiation were 267 (IQR 175–402.5) cells/mm³ and 37 593 (IQR 6870.50–194 876.50) copies/mL, respectively. Four patients had TB at the start of treatment.

At the time of the start of this study, the initial regimen for third-line therapy was raltegravir (400 mg)+darunavir (600 mg)+ritonavir (100 mg); given as one tablet twice daily. In

Table 2. Baseline drug resistance mutations against PR, RT & IN in patients who died during third-line treatment

Patient	Mutations at start of third-line therapy						Resistance against PR and IN	Regimen at second-line failure	VL at 12 months (copies/mL)	Time of death from initiation of third-line (months)	Cause of death
	PR		RT		IN						
	Major	Accessory	NNRTI	NRTI	Major	Accessory					
1	M46I, I50L	F53L, Q58E, G73S	V106M, V108I, Y181C	M41L, E44D, D67N, M184V, T215Y	Y143C	None	ATV/r ^H , CAB ^{PL} , EVG ^{PL} , RAL ^H	TL + ATV/r	TND	31	Stopped therapy
2	M46I, N88S	K20T, F53L, G73S	A98G, K103N, V108I	M41L, E44D, D67N, T69D, V75M, M184, L210W, T215Y, K219N	NA	NA	ATV/r ^H , LPV/r ^{LL}	ABCL + LPV/r	227	23	Chicken pox
3 ^b	None	None	None	None	E92DV	None	CAB ^{PL} , EVG ^H , RAL ^I	TL + ATV/r	89 862	49	TB
4	N88T	None	K103N, E138Q	M184V, T215C	NA	NA	ATV/r ^{LL}	TL + ATV/r		11 ^a	Fever
5	M46I	L89T	A98G, E138Q, G190A	M41L, E44D, M184V, T215Y	None	None	ATV/r ^I , LPV/r ^I	ZL + ATV/r		1 ^a	TB
6	None	I47T	None	None	None	None		TL + ATV/r		5 ^a	Stopped therapy
7 ^b	None	None	None	None	None	None	BIC ^I , CAB ^I , DTG ^I , EVG ^H , RAL ^H	TL + ATV/r	1 768 624	18	Unknown
8	None	None	K101E, G190A	None	Q148K	V151A		ZL + ATV/r	203	44	Fever
9	NA	NA	NA	NA	Y143N, N155H	E157Q	BIC ^{PL} , CAB ^{LL} , DTG ^{PL} , EVG ^H , RAL ^H	TL + ATV/r		2 ^a	Unknown
10	N88S, L90M	K20T	Y188L	M41L, E44D, M184V, L210W, T215Y	None	None	ATV/r ^H , LPV/r ^{LL}	TL + ATV/r		8 ^a	TB
11	NA	NA	NA	NA	None	None		TL + ATV/r		27	COVID-19
12	None	None	None	None	E92D, Y143C	None	CAB ^{PL} , EVG ^{PL} , RAL ^H	ZL + ATV/r	TND	24	Unknown
13	NA	NA	K103N, V108I	M41FL, K65N, D67E, L74I, M184V, L210W, T215Y	None	None		ZL + ATV/r		11 ^a	Stopped therapy

^aPatients died within a year.

^bDetail of mutations at the time of virological failure are given in Table 3. H, high-level resistance; I, intermediate-level resistance; PL, potential low-level resistance; LL, low-level resistance; BIC, bictegravir; CAB, cabotegravir; DTG, dolutegravir; EVG, elvitegravir; RAL, raltegravir; ATV/r, atazanavir; LPV/r, lopinavir; TL, tenofovir and lamivudine; ABCL, abacavir and lamivudine; ZL, zidovudine and lamivudine; NA, not available; TND, target not detected.

May 2020, the national programme replaced raltegravir with dolutegravir for all patients.

Survival outcomes

At 12 month follow-up of third-line therapy, 72/85 (85%) patients were alive on ART, 6/85 (7%) patients died, and 7/85 (8%) were LFU. At the end of follow-up in March 2022, 61/85 (72%) patients were alive and on ART, which included 1 patient who was LFU after 1 year of therapy and later restarted therapy. Thirteen of the 85 patients (15%) died during third-line therapy. (Figure 1) Eleven of the 85 patients (13%) were LFU at the end of follow-up of the study (Figure 1). Out of the seven patients who died beyond 1 year, only two had virological failure (Table 2).

Virological outcomes

Out of 72 patients who were alive at 12 month follow-up, 59/72 (82%) patients had suppressed VL (<150 copies/ml), while 13/72 (18%) did not attain virological suppression. At the end of follow-up of the study in March 2022, VL could be done in 67 patients, out of which 59 (88%) had virological suppression. Five out of 13 patients (39%) who failed therapy at 12 months were virologically suppressed at the end of follow-up. One patient who restarted therapy after 1 year did not attain virological suppression at the end of follow-up of the study (Figure 2).

HIV DRMs

Out of 85 patients started on third-line therapy, 40 samples were successfully sequenced for the IN gene and 38 for the PR and RT genes at baseline (start of third-line therapy). All specimens were subtype C, except two, which were subtype B. At baseline, 35% (14/40) of these patients had major mutations associated with the IN gene: [E92D ($n=5$); T66P ($n=2$); Y143C/S/N ($n=7$); N155H/T ($n=4$); Q148K/L ($n=2$); G140A ($n=1$); and G118R ($n=1$)], while 7 had accessory IN mutations. Out of these 14 patients, 5 died, 8 patients were still on ART and virologically suppressed and 1 patient was LFU in 2022. Seventeen out of 38 (45%) patients had major PI mutations (with a higher frequency of N88S, M46I and L90M) at baseline, out of which 5 patients

died, 1 patient was LFU and 11 patients were virologically suppressed and on ART at the end of follow-up. NRTI- and NNRTI-associated mutations were found in 25/38 (66%) and 28/38 (74%) patients, respectively.

Among the 13 patients who did not attain virological suppression at 12 months, genotyping could be done in only 12. At the initiation of third-line therapy among these 12 samples, 10 samples were successfully sequenced for the IN gene and 11 were sequenced for PR and RT genes. Details of DRMs at baseline and during virological failure are shown in Table 3. After 1 year of follow-up, out of these 12 samples of virological failure patients, 11 were successfully sequenced for the IN gene and 10 were sequenced for PR and RT genes. Four patients (4/11; 36%) had major IN mutations and 3/11 (27%) patients had IN accessory mutations. We did not detect any major PI mutations, while PI accessory mutations were found in 1/10 (10%) patient only (Table 3).

Details of the DRMs of the 13 patients who died during treatment are given in Table 2. At baseline, out of these 13 samples, 11 were sequenced for the IN gene and 10 samples were sequenced for PR and RT genes. Five (5/11; 46%) patients had major IN mutations and 2/11 (18%) patients had IN accessory mutations. Five (5/10; 50%) patients had major PI mutations and 5/10 (50%) patients had PI accessory mutations. Six patients had NRTI mutations and seven patients had NNRTI mutations.

Discussion

This is the first study from India that has observed the long-term outcome of third-line ART (median follow-up was 42 months) of patients in programmatic settings and mutational analysis in those failing treatment at 1 year. The survival of our patients at 1 year was 85%. This was similar to third-line treatment studies done under programmatic conditions from India (81%),⁹ Zimbabwe (79%)¹³ and South Africa (83%),⁶ while it was slightly lower than that of a private clinic in western India.¹⁴ At the end of follow-up in March 2022, 15% of our patients had died. Although there is no study on long-term outcome of third-line ART in real-time settings, this was very high compared with 4% death,

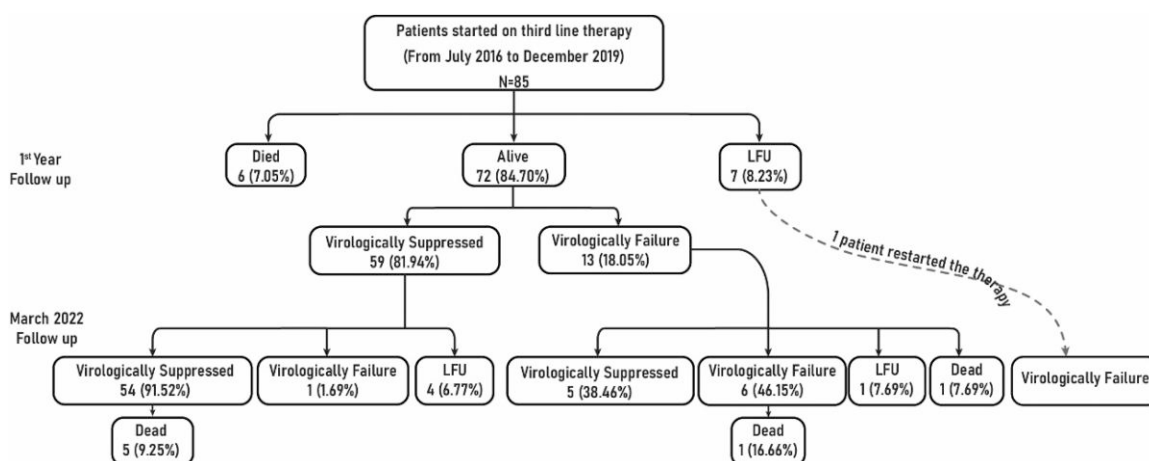


Figure 2. Virological outcomes of patients started on third-line therapy.

Table 3. DRMs at baseline and at the time of failure in patients with virological failure at 1 year

Patient	Mutations at baseline				Mutations at virological failure (at 12 months)				Resistance level against PR and IN at baseline	Resistance level against PR and IN at 12 months	Regimen at second-line failure	Current VL (copies/mL)	Current status																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																			
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H, high-level resistance; I, intermediate-level resistance; PL, potential low-level resistance; LL, low-level resistance; BIC, bictegravir; CAB, cabotegravir; DTG, dolutegravir; EVG, elvitegravir; RAL, raltegravir; ATV/r, atazanavir/r; LPV/r, lopinavir/r; TL, tenofovir and lamivudine; ZL, zidovudine and lamivudine; NA, not available; TND, target not detected.

^aDied during treatment.

observed in the ACTG A5288 clinical trial done in low- and middle-income countries.¹⁵ However, it was encouraging to observe that in our own centre, survival at the end of follow-up in highly treatment-experienced patients on third-line ART (72%) was better than those on second-line therapy (65%).¹⁶ In our third-line cohort, 46% (6/13) of the deaths occurred within 12 months; beyond that, among the 7 patients who died only 2 had virological failure, suggesting the treatment was still working for most PLHIV. Risk factors for death were poor WHO clinical stage and low weight at the time of starting third-line therapy, which suggests delayed diagnosis of second-line treatment failure. Baseline mutational analysis among those who died showed that 50% (5/10) had major PI mutations, while 45% (5/11) had major IN mutations, suggesting that the patients may have been on a failing regimen for a long time before third-line therapy was initiated. Thus, the best way to improve survival is by detecting patients with second-line failure and initiating third-line therapy early. This can be done by ensuring ART centres do regular VL monitoring as per guidelines and early referral to higher centres for initiation of third-line therapy. As all the third-line patients have to be referred to CoEs or ART Plus centres for evaluation. Efforts should also be made to make the referral process of these patients more efficient. This can be done easily by using virtual platforms for referring patients.

Retention in care is another critical component of HIV care interventions and the 12% LFU in the cohort needs to be addressed for the success of the programme. Out of the eleven patients who were LFU, 7 were from the neighbouring state of Bihar and rest were from other districts of Uttar Pradesh. One of the main reasons for high LFU may be the fact that patients from distant areas come for third-line ART to our centre, as third-line drugs are only available through the CoEs and ART Plus centres in the programme. These centres may be far from a patient's residence, therefore improving accessibility of these drugs would definitely improve retention in care.

At 1 year follow-up, 82% of patients (59/72) in our study attained virological suppression, similar to studies from both India^{4,9} and other low- and middle-income countries.^{6,13} VL suppression was maintained at the end of follow-up of the study in 88% of patients (59/67). This finding is similar to the ACTG A5288 trial, where 83% and 80% of the participants had VL suppression at Weeks 144 and 168, where the patients were not given a standardized third-line regimen like our setup but antiretrovirals were based on their genotyping results. This suggests that standardized robust regimens for highly treatment-experienced PLHIV in public health programmes are as effective as regimens guided by genotyping results.

Surprisingly, 5 out of 13 patients who had virological failure at 12 months showed VL suppression at the end of the study. One reason could be an adherence issue as four of these five patients showed no major PI or IN mutations at the time of failure. Another reason for virological suppression could be the fact that there was a change in integrase inhibitor in the programme from raltegravir to dolutegravir in 2020 and dolutegravir is known to be effective, even in patients with raltegravir failure.

At the start of the third-line regimen, 35% (14/40) had major INSTI DRMs, even when they had never been on an INSTI-based regimen, similar to the study by Gill *et al.*¹⁴ As the national

programme has adopted dolutegravir-based regimens as first-line treatment in 2020, the presence of INSTI DRMs in INSTI-naïve PLHIV is concerning and should be explored further. Among those who had virological failure at 1 year, 33% (4/12) patients had major IN mutations but we did not detect any major PI mutations. None of the patients had resistance against darunavir and only two had resistance against dolutegravir. There are no real-world data of DRMs acquired in patients with third-line failure in the programme. However, in contrast to our findings, in a study conducted at a tertiary care private HIV clinic in Pune, India, 90% of PLHIV failing raltegravir-based ART had at least one IN resistance associated mutation.¹⁷ The INSTI DRMs identified in our study both at baseline and at treatment failure were similar to this study. The low prevalence of DRMs, even after failing third-line therapy, bodes well for the programme. It reiterates the fact that the standardized third-line antiretrovirals provided free in the programme are highly effective.

The limitation of our study is that data are from a single centre; however, as standardized ART is given throughout the programme, overall outcome should not be different from this study.

To conclude, this study shows good long-term treatment outcome for patients on standardized third-line therapy in programmatic conditions. To further improve outcomes, priority needs to be given to early detection of treatment failure and retention in care in the programme. Timely VL monitoring, along with decentralization and easy accessibility of third-line drugs in the peripheral ART centres, will definitely decrease death and LFU patients on third-line ART in the programme.

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Transparency declarations

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